GEOPOLITICAL RISK, CAPITAL FLOW VOLATILITY, AND ASSET MARKET SPILLOVERS

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NO. 820

November 2025

ADB ECONOMICS WORKING PAPER SERIES

ASIAN DEVELOPMENT BANK

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ADB Economics Working Paper Series

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ISSN 2313-6537 (print), 2313-6545 (PDF)
Publication Stock No. WPS250470-2
DOI: http://dx.doi.org/10.22617/WPS250470-2

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ABSTRACT

This paper examines the effects of country-specific geopolitical risk on capital flow volatility and asset markets across 29 emerging and advanced economies over the period 2000–2023. Using panel regressions and a panel structural vector autoregression framework, the results show that geopolitical risk raises bond yields and leads to exchange rate depreciation, with stronger and more persistent effects in emerging economies. Asset markets for advanced economies are affected mainly through lower equity prices. The impact on capital flow volatility is slightly higher on average for advanced economies but remains more persistent for emerging economies. Greater financial development, higher central bank independence, and lower public debt mitigate the adverse effects of geopolitical risk on both capital flows and asset markets. These findings highlight the importance of strong macroeconomic fundamentals and institutional frameworks in building resilience against geopolitical shocks.

Keywords: geopolitical risk, capital flow volatility, financial markets

JEL codes: G15, G41

I. INTRODUCTION

Geopolitical risk has emerged as a major factor affecting global financial markets and global economic prospects. Episodes of amplified geopolitical tensions can trigger rises in global risk aversion and spikes in oil prices, with spillovers to asset prices and the trajectory of capital flows. Recent work in this growing literature has highlighted that the key channels through which geopolitical risk affects financial markets are investor sentiment (e.g., Hachicha 2023, Banerjee, Kumar, and Goodell 2024) and market liquidity (e.g., Abdel-Latif and El-Gamal 2020), with emerging markets particularly affected (e.g., Das, Kannadhasan, and Bhattacharyya 2019; Mansour-Ichrakieh and Zeaiter 2019).

While the literature has tended to focus on the macroeconomic consequences of geopolitical risks, there has been less work examining the spillover effects of geopolitical risks on capital flow volatility and asset markets, especially from the comparative perspective of emerging and advanced economies. A few recent studies have explored the negative impact of global geopolitical risk on global financial markets (Baur and Smales 2020, Smales 2021, Izzeldin et al. 2023, Yilmazkuday 2024). However, there is less direct empirical evidence on the effects of country-specific geopolitical risks on domestic financial markets and capital flow volatility. Against this backdrop, this paper employs a set of panel analyses, as well as a panel structural vector autoregression (VAR) model, to estimate the impacts of geopolitical risk on domestic asset markets and capital flow volatility in 29 emerging and advanced economies over the period 2000 to 2023.

To measure the country-specific geopolitical risks, we directly use the monthly country-specific geopolitical risk index (GPR index) constructed by Caldara and Iacoviello (2022), which offers two advantages for studying the relationship between geopolitical risk and financial markets. First, unlike data that would inform only sporadic geopolitical events, the GPR index provides monthly data dating back to 1985, offering a long time series that allows for an in-depth analysis of capital flow and asset price dynamics in response to GPR fluctuations. Second, unlike many other uncertainty proxies, the GPR index does not consistently rise during recessions or financial crises but spikes in response to events like wars or terrorist attacks. This makes the GPR index particularly useful for isolating the effects of geopolitical risk on financial markets posed by other forms of uncertainty and for addressing endogeneity concerns.

Our analyses consist of two stages. The first stage uses a panel regression framework to examine the role of geopolitical risks in affecting capital flow volatility and asset markets, controlling for both domestic fundamentals and global factors. In the case of emerging economies, geopolitical risk has a significant positive effect on bond yields, while equities significantly decline and these economies also incur significant currency depreciation. The effects on bond yields and currencies are stronger for emerging than for advanced economies. However, for equity prices, the impact is marginally higher in magnitude in the case of advanced economies. In addition, as regards capital flow volatility, we find that the average impact is slightly higher for advanced economies.

To explore whether the heterogeneity in country-specific characteristics across countries could affect domestic asset market responses to geopolitical risks, we estimate the panel regressions' conditioning on financial development, central bank independence, and public debt levels. These conditioning factors are shown to have a moderating effect on the impact of geopolitical risks on asset markets and capital flow volatility. Countries with higher financial development exhibit weaker responses in stock prices, bond yields, and exchange rates, though they remain more vulnerable to capital flow volatility owing to their greater global financial integration. Conversely, economies with lower financial development experience more severe depreciation in exchange rates. Similarly, countries with greater central bank independence show resilience to geopolitical

risks, with the adverse effects on domestic asset markets yielding insignificant results. In contrast, countries with lower central bank independence face sharper declines in stock prices and exchange rates alongside rising bond yields and capital flow volatility. Public debt also plays a critical role, with lower-debt countries experiencing milder financial market disruptions, while high-debt economies face amplified adverse effects.

Finally, we investigate the dynamic effects of geopolitical risks on capital flow volatility and financial market variables using a panel structural VAR (SVAR) model. This approach, which also controls for domestic fundamentals and global factors identified in the baseline panel ordinary least squares (OLS) model, allows us to capture both the duration and the dynamics of these effects over time. These dynamic results provide further evidence that heightened geopolitical risks significantly increase capital flow volatility and long-term bond yields, exacerbating financial instability. Geopolitical risk shocks also exert significant and persistent negative effects on stock prices and exchange rates, indicating that rising geopolitical risks weaken both equity and currency markets. In the case of emerging economies, the response of capital flow volatility to geopolitical risk shocks is higher in magnitude at peak than that of advanced economies, while the response also remains statistically significant for a longer duration (for around 6 months, compared to 3 months for advanced economies). The dynamic analysis also reveals a stronger worsening reaction in bond yields, stock prices, and currencies in the case of emerging compared to advanced economies.

Overall, we find that heightened geopolitical risks disrupt domestic asset markets and increase capital flow volatility. The severity of these adverse effects varies depending on country-specific characteristics. Our findings suggest that greater central bank independence and lower debt can help mitigate the impact of geopolitical risks on asset markets and capital flow volatility. Enhancing financial development can also help moderate the effects of geopolitical risks on asset markets. However, effective macroprudential policy and capital flow management frameworks are key aspects for consideration in helping mitigate capital flow volatility. In addition, given that rising geopolitical risks can create a more challenging environment for central banks to implement effective monetary policies, this may have implications for monetary policy operational frameworks. Policymakers should also strengthen coordination between monetary and macroprudential policies, which can help stabilize markets during episodes of elevated geopolitical risk.

This paper contributes to the growing literature on the financial market implications of geopolitical risk. The existing literature on the financial market implications of geopolitical risks tends to focus on particular asset market segments, particularly the stock market and commodities. Balcilar et al. (2018) find that geopolitical risk largely affects stock market volatility rather than returns across the BRICS economies.[1] Similar effects are found by Salisu et al. (2022) and Zhang et al. (2023) for other sets of economies. Hoque and Zaidi (2020) note that significant nonlinearity prevails in the relationship between geopolitical risk and emerging stock markets.

Some other studies have focused on the effects on particular sectors, including defense, tourism, and renewable energy (e.g., Apergis et al. 2018, Demiralay and Kilincarslan 2019, Jiang et al. 2020). On commodities, several studies have examined the role of geopolitical risk in amplifying commodity price volatilities in respect of oil (e.g., Liu et al. 2019; Mei et al. 2020; Qin et al. 2020; Wang, Su, and Umar 2021; Behera, Priyadarsini, and Patnaik 2024) and precious metals (e.g., Banerjee, Ghosal, and Mukherjee 2019; Gkillas, Gupta, and Pierdzioch 2020).

[1] BRICS refers to Brazil, Russia, India, the People's Republic of China, and South Africa.

Some recent work has explored its implications for currencies and capital flows. Ma (2025) finds that heightened geopolitical risk leads to currency depreciation in Asia, while Zhang and Chen (2025) show that rising economic uncertainty increases capital flow volatility.

However, the broader question of how geopolitical risk affects capital flow volatility and asset market returns across both emerging and advanced economies remains largely unexplored. Moreover, the literature offers limited insights into the role of domestic fundamentals in mitigating such spillovers.

This paper fills these gaps in three ways. First, we provide a comprehensive cross-country analysis of the effects of geopolitical risk on both capital flow volatility and three asset market returns across 29 advanced and emerging economies from 2000 to 2023. Second, we identify the role of domestic fundamentals (notably financial development, central bank independence, and public debt levels) in mitigating the extent of geopolitical risk spillovers. Third, by using both panel regression and panel SVAR approaches, we capture not only the average effects of geopolitical risk but also its dynamic propagation over time.

The remainder of this paper is structured as follows. Section II describes the data and outlines the empirical methodology. Section III presents the empirical results. Section IV concludes.

II. DATA AND EMPIRICAL METHODOLOGY

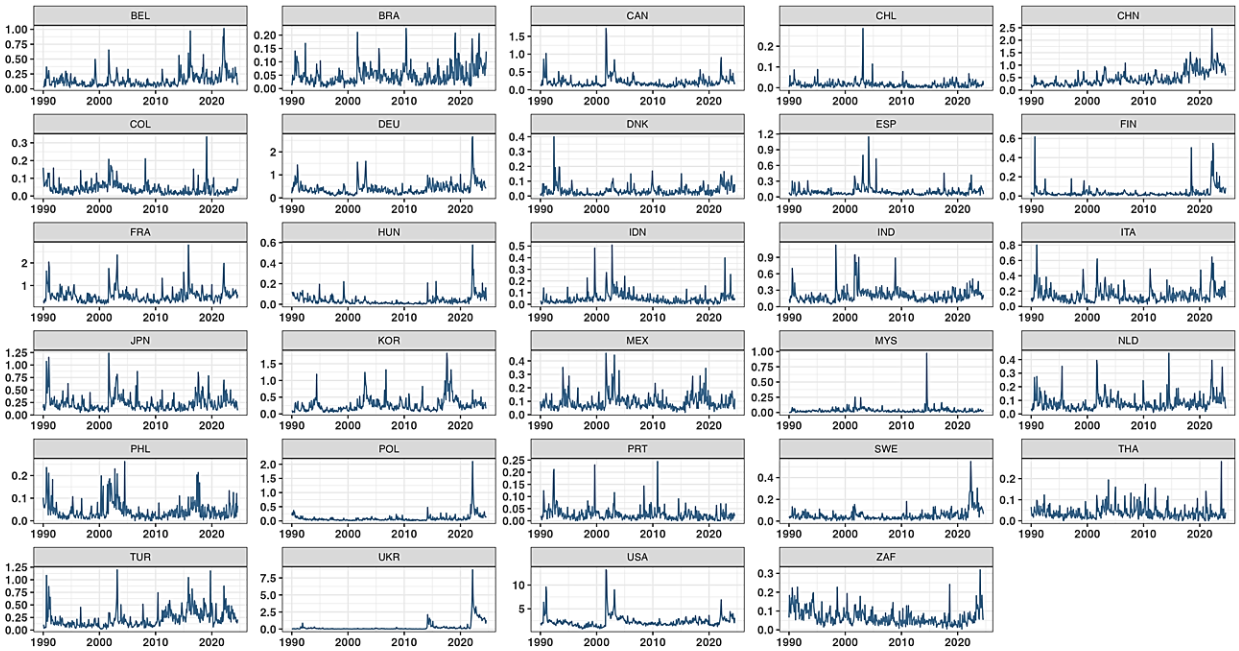
In this section, we first discuss the measure of geopolitical risk. We then describe the data source for variables we use in the empirical analysis. Finally, we present our econometric framework used to produce the empirical results.

A. Country-Specific Measures of Geopolitical Risk

Geopolitical risk (GPR) is defined as the uncertainty arising from wars, terrorist acts, and interstate tensions that disrupt the normal functioning of international relations. In this paper, we use the monthly country-specific GPR index constructed by Caldara and Iacoviello (2022), which follows the methodological frameworks developed by Saiz and Simonsohn (2013) and Baker, Bloom, and Davies (2016). The index is based on the frequency of articles referencing geopolitical events and risks in 11 major English-language newspapers: The Boston Globe, The Chicago Tribune, The Daily Telegraph, Financial Times, The Globe and Mail, The Guardian, Los Angeles Times, The New York Times, The Times, The Wall Street Journal, and The Washington Post. The GPR index is computed as the ratio of geopolitical articles to total articles.[2] Figure 1 plots country-specific GPR indices for our sample countries.

[2] Specifically, Caldara and Iacoviello (2022) construct country-specific measures of geopolitical risk by counting joint occurrences in newspapers of geopolitical terms and the name of the country (or its capital or main city) in question. By construction, the severity of a particular geopolitical shock cannot be ascertained, however.

Figure 1: Country-Specific Geopolitical Risk



Notes: The figure plots the dynamic of country-specific geopolitical risks for the sample of countries, obtained from Caldara and Iacoviello (2022). For each country, the country-specific GPR index measures the share of articles simultaneously mentioning geopolitical risks together with the name of the country (or its capital or main city) in question. The indicators BEL, BRA, CAN, CHL, CHN, COL, DEU, DNK, ESP, FIN, FRA, HUN, IDN, IND, ITA, JPN, KOR, MEX, MYS, NLD, PHL, POL, PRT, SWE, THA, TUR, UKR, USA, and ZAF represent Belgium, Brazil, Canada, Chile, the People's Republic of China, Colombia, Germany, Denmark, Spain, Finland, France, Hungary, Indonesia, India, Italy, Japan, the Republic of Korea, Mexico, Malaysia, the Netherlands, the Philippines, Poland, Portugal, Sweden, Thailand, Türkiye, Ukraine, the United States, and South Africa, respectively.

Source: Authors' calculations.

B. Data

We use available monthly data with an unbalanced panel for 29 countries, covering both emerging market economies and advanced economies, spanning 2000:M1 to 2023:M12.[3] We collect data on the real stock price index and 10-year government bond yield as indicators of the domestic financial market. For the exchange rate, we use the real effective exchange rate index. The data are from the Organisation for Economic Co-operation and Development (OECD) Main Economic Indicators, the International Monetary Fund (IMF) International Financial Statistics, and the Bank for International Settlements (BIS) database.

For capital flows, we utilize the OECD Monthly Capital Flow Dataset (de Crescenzio and Lepers 2024), which offers several advantages over private data providers such as Emerging Portfolio Fund Research (EPFR) and the Institute of International Finance (IIF). First, the OECD dataset

[3] The selected economies are Belgium, Brazil, Canada, Chile, the People's Republic of China, Colombia, Germany, Denmark, Spain, Finland, France, Hungary, Indonesia, India, Italy, Japan, the Republic of Korea, Mexico, Malaysia, the Netherlands, the Philippines, Poland, Portugal, Sweden, Thailand, Türkiye, Ukraine, the United States, and South Africa. The data starting year for a specific variable varies across countries depending on data availability and reliability.

provides more comprehensive country coverage, including both emerging market economies and advanced economies, offering a more global perspective on capital flows. Second, it includes a wider range of financial transactions, such as bank flows and foreign direct investment (FDI), and captures both inflows and outflows, facilitating a comprehensive analysis of the financial account. Third, the dataset is aligned with balance of payments (BoP) principles, enhancing its reliability for cross-country economic comparisons. Fourth, the OECD dataset is publicly available, reducing dependency on costly private data sources, and is updated monthly, providing timely insights consistent with BoP standards. Specifically, we focus on portfolio inflows, which constitute a significant portion of the dataset, enabling us to extend the time series for a broader range of countries. To measure capital flow volatility, we calculate the standard deviation of capital flows using a 3-month rolling window of monthly data.[4]

To assess the impact of geopolitical risk on capital flow volatility and financial markets, we incorporate several control variables for both domestic fundamentals and global factors, based on the related literature. For domestic macroeconomic fundamentals, we include the real industrial production index as a measure of domestic output, year-on-year changes in the consumer price index as an indicator of inflation, and central bank policy rates. These data are sourced from the IMF's International Financial Statistics and the BIS database. For global factors, we use data from various sources: the West Texas Intermediate (WTI) crude oil price as a proxy for global oil prices, the VIX index (the Chicago Board Options Exchange [CBOE] Volatility Index) to capture global financial market uncertainty, and the Economic Policy Uncertainty (EPU) index from Baker, Bloom, and Davies (2016) to measure global economic policy uncertainty.

Recognizing the importance of structural country-specific characteristics, we include several additional variables in our empirical analysis. To gauge the level of financial development, we use the IMF's Financial Development Index, which assesses the depth, access, and efficiency of financial institutions and markets. We also incorporate the Central Bank Independence Index from Romelli (2022) as a measure of the degree of central bank independence. Finally, we include the government debt-to-gross domestic product (GDP) ratio, sourced from the IMF's World Economic Outlook, to capture public indebtedness.

Table 1 presents the summary statistics of the main variables used in the empirical analysis.

[4] Given the lack of monthly gross domestic product (GDP) data for our sample countries, and recognizing that weighting capital flows with quarterly GDP may excessively smooth volatility (Broto, Díaz-Cassou, and Erce 2011), we refrain from using a ratio form to represent the capital flow variable. We chose a 3-month rolling window to better capture the effect of geopolitical risks on short-term dynamics, given the high-frequency nature of changes in geopolitical risks. To ensure the robustness of our findings, we tested the estimates using alternative rolling window lengths, such as 6 months and 12 months. Additionally, we employed alternative measures of capital flow volatility, including GARCH(1,1) and ARIMA(p,1,q) models, and found that our results remained consistent and robust across these specifications.

Table 1: Summary Statistics

Variable	Mean	SD	Min.	Max.
Capital flow volatility (SD)	7.22	19.62	0	528.40
Real stock prices (log)	4.32	0.47	2.37	5.46
Long-term bond yield (%)	4.74	3.63	-0.65	42.18
Real effective exchange rate (log)	4.64	0.18	3.61	5.23
Geopolitical risk (index)	0.22	0.52	0	13.23
Policy rate (%)	4.37	4.87	-0.75	57
Real industrial production (log)	4.58	0.31	2.29	5.81
Inflation (%)	4.06	6.18	-4.43	85.51
Oil price (log)	3.73	0.63	2.42	4.90
VIX (log)	19.59	7.54	10.13	62.67
Global economic policy uncertainty (index)	140.26	74.91	49.23	431.73
Financial development (index)	0.60	0.19	0.11	0.93
Central bank independence (index)	0.66	0.19	0.14	0.93
Public debt to GDP (%)	68.77	36.40	11.82	216.33

GDP = gross domestic product, SD = standard deviation, VIX = Chicago Board Options Exchange Volatility Index.

Source: Authors' calculations.

C. Econometric Methodology

Our econometric methodology is underpinned by international portfolio choice and asset pricing theories, augmented with a geopolitical risk component. Theory indicates that higher uncertainty driven by elevated geopolitical risk triggers capital flow volatility as investors rebalance toward safe havens. Meanwhile, equity returns decline, sovereign risk premia rise, and currencies face depreciating pressures. More specifically, our empirical framework is founded on a generalized capital asset pricing model (Sharpe 1964) of the following form:

$$E(R)_i = R_f + \beta_i\left[E(R)_m - R_f\right] + \lambda_i(Risk_i) \quad (1)$$

where $E(R)_i$ is the expected return on asset *i*, R_f is the risk-free rate, β_i is the asset's exposure to the global markets, and λ_i denotes exposure to country-specific risk factors, including geopolitical risk.

Based on the Capital Asset Pricing Model framework, priors are informed on how heightened geopolitical risk can be expected to affect capital flow volatility and asset markets. For example, where geopolitical risk rises, the required return on assets would rise in order to compensate investors. This would trigger an outward reallocation of capital and amplified capital flow volatility. The same underlying premise applies to equity prices, bond yields, and exchange rates. More specifically, higher geopolitical risk leads to a higher risk premium on assets and a higher required return for holding that asset, triggering reallocation to less risky assets. Higher required returns on equities imply lower equity prices, higher required returns on bonds imply lower bond prices (or higher yields to compensate for the higher risk of default), and currencies depreciate as a higher risk premium leads to safe-haven flows of capital out of risky assets.

Turning to the methodology, the first stage examines the role of geopolitical risks in affecting capital flow volatility and domestic asset markets, controlling for both domestic fundamentals and global factors. Specifically, the following baseline equation is estimated:

$$y_{i,t}^{j} = \beta GPR_{i,t} + \gamma X_{i,t-1} + GEPU_{t-1} + VIX_{t-1} + \delta_i + \varepsilon_{i,t} \quad (2)$$

where $i = 1, \cdots N$ refers to the specific economy in the sample; y is the variable of interest,

including capital flow volatility, the log of stock prices, long-term bond yield, and the log of the real effective exchange rate; GPR is the series of the GPR index (standardized by country); and X is a vector of domestic fundamentals, including the log of real industrial production, inflation, and the policy rate, along with country-specific characteristics such as financial development, central bank independence, and the public debt-to-GDP ratio. $GEPU$ and VIX represent the log of the global EPU index and the log of the VIX index, respectively, to account for global factors. δ_i denotes economy-specific fixed effects, controlling for the time-invariant characteristics of the country. Finally, $\varepsilon_{i,t}$ denotes error term. Control variables are lagged by one period to mitigate potential endogeneity concerns.

To examine whether heterogeneity in country-specific characteristics influences the effect of geopolitical risk, we divide the sample into groups based on whether they are emerging or advanced economies, their levels of financial development, central bank independence, and public indebtedness, and estimate separate regressions for each group.

We further examine the dynamic responses of capital flow volatility and financial market variables to geopolitical risk shocks using a panel SVAR model. The analysis also controls for the domestic fundamentals and global factors outlined in the baseline OLS model. This approach enables us to identify the dynamics and duration of the effect of geopolitical risks. The panel SVAR can be denoted as follows in its general specification, with structural shocks identified by a recursive restriction:

$$Y_{i,t} = A(L)Y_{i,t-1} + \alpha_i + \mu_{i,t} \tag{3}$$

where $Y_{i,t}$ refers to a vector of our selected endogenous variables of country i, including the GPR index, capital flow volatility, the logarithm of stock prices, the long-term bond yield, and the logarithm of the real effective exchange rate. Additionally, global factors are represented by the logarithm of the global EPU index and the logarithm of the VIX index, while domestic fundamentals include the logarithm of real GDP, the inflation rate, and the policy rate; $A(L)$ is a matrix of polynomials in the lag operator L; α_i denotes country-specific fixed effects to account for unobserved time-invariant heterogeneity across countries; and μ_t is a vector of disturbances. Given the autoregressive nature of the panel SVAR model, fixed effects are intrinsically correlated with the regressors. Hence, we use the forward orthogonal deviation procedure to eliminate fixed effects, such that the transformed variables are orthogonal to the lagged regressors. The lag structure in the panel SVAR (three lags) has been selected using the Schwarz information criterion (SIC).

To isolate the impact of geopolitical risk from the current or past performance of domestic fundamentals and global factors, we adopt a recursive identification scheme as outlined by Christiano, Eichenbaum, and Evans (1999). In this framework, variables ordered higher in the sequence are insulated from contemporaneous shocks originating from those ordered below them, while lower-ranked variables are influenced by contemporaneous shocks from those above. Specifically, we position the GPR index at the top of the ordering, ensuring it is influenced only by its own contemporaneous shocks.[5] Following this, we place the global factor variables, meaning they are affected by both their own shocks and those from the GPR index, but remain unaffected by any contemporaneous shocks from variables ordered below them. Next, we order the domestic macroeconomic fundamentals, ensuring they are placed before the capital flow volatility and financial market variables. Importantly, we position capital flow volatility and financial market variables at the bottom of the ordering.

This ordering scheme is based not only on the assumption that these variables are influenced by

[5] This assumption is supported by the characteristics of the GPR index, as outlined by Caldara and Iacoviello (2022).

the GPR index, global factors, and domestic fundamentals but also on our first-stage empirical results, which indicate that capital flow volatility and financial markets are driven by both push and pull factors. The resulting relationships between the variables within the SVAR setup are therefore manifested in the impulse response functions. As a robustness test on the impulse response functions, we also examine an alternative ordering scheme in the SVAR.

III. EMPIRICAL RESULTS

A. Baseline Results

Table 2 presents the estimated impact of geopolitical risks on capital flow volatility, stock prices, bond yields, and exchange rates, controlling for domestic fundamentals, global factors, and country-specific characteristics as specified in Equation (2). The results show that the GPR index significantly increases capital flow volatility (column 1) and long-term bond yields (column 3), indicating that heightened geopolitical risk exacerbates capital flow volatility and drives bond yields higher. Additionally, the GPR index has a significant negative effect on stock prices (column 2), implying that rising geopolitical risks depress equity markets. The impact on exchange rates for the full sample is not significant, however.

Our model accounts for a variety of domestic and global factors, the effects of which align with the related literature. For instance, higher real output and inflation, coupled with a lower policy rate, increase stock prices while reducing bond yields. Global factors, such as rising oil prices, VIX, and the global EPU index, play a pivotal role in driving capital flow volatility, as increases in these factors significantly raise volatility levels. Fiscal sustainability is also crucial: worsening fiscal conditions lead to increased capital flow volatility and lower stock prices, emphasizing the importance of sound fiscal policies in mitigating the adverse effects of geopolitical risks.

Table 2: Geopolitical Risks, Capital Flow Volatility, and Financial Markets—Baseline Panel Estimates

	(1) Capital Flow Volatility	(2) Stock Prices	(3) Bond Yield	(4) Exchange Rate
GPR index	0.867**	-0.039***	0.134***	0.001
	(0.382)	(0.006)	(0.030)	(0.002)
	Domestic fundamentals			
Policy rate	0.175	-0.041***	0.598***	-0.007***
	(0.155)	(0.002)	(0.012)	(0.001)
Real industrial production	2.582	0.289***	-1.671***	0.374***
	(1.926)	(0.021)	(0.132)	(0.007)
Inflation	0.075	0.014***	-0.117***	-0.007***
	(0.074)	(0.001)	(0.009)	(0.001)
	Global factors			
Oil prices	4.824***	0.023***	0.233***	0.081***
	(0.700)	(0.009)	(0.051)	(0.003)
VIX	1.836**	-0.271***	1.098***	0.041***
	(0.831)	(0.011)	(0.061)	(0.004)
Global EPU index	3.107***	0.144***	-1.411***	-0.017***
	(0.833)	(0.011)	(0.060)	(0.004)

Continued on the next page

	(1) Capital Flow Volatility	(2) Stock Prices	(3) Bond Yield	(4) Exchange Rate
Country-specific characteristics				
Financial development	0.055	0.013***	0.000	-0.002***
	(0.042)	(0.001)	(0.003)	(0.000)
Central bank independence	-0.047	0.000	0.005***	0.000
	(0.030)	(0.000)	(0.002)	(0.000)
Public debt/GDP	0.212***	-0.006***	0.001	-0.000***
	(0.021)	(0.000)	(0.002)	(0.000)
Observations	4,779	5,671	5,412	5,671
R-squared	0.058	0.385	0.556	0.517
Country fixed effects	YES	YES	YES	YES

EPU = Economic Policy Uncertainty, GDP = gross domestic product, GPR = geopolitical risk, VIX = Chicago Board Options Exchange Volatility Index.

Notes: The table presents the estimated impact of geopolitical risks on capital flow volatility, stock prices, bond yields, and exchange rates. Control variables include domestic fundamentals, global factors, and country-specific characteristics as specified in Equation (2). Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Source: Authors' calculations.

B. Heterogeneity Analyses

In this part of the analysis, we explore how the impact of geopolitical risks on capital flow volatility and asset markets varies based on country-specific characteristics. Specifically, we examine regional differences (i.e., emerging versus advanced economies), the effects conditional on the extent of financial development, central bank independence, and public debt levels. By comparing these factors across countries, we gain insights into how different structural and macroeconomic environments influence the transmission of geopolitical shocks into financial markets.

1. Emerging Versus Advanced Economies

We further disaggregate our sample into emerging and advanced economies to explore regional heterogeneity in the effects of geopolitical risks. Table 3 presents the estimated impact of these risks on capital flow volatility, stock prices, bond yields, and exchange rates across these regions. The results underscore notable differences between the two groups. For emerging economies, the effects of heightened geopolitical risks on bond yields and exchange rates are particularly pronounced when these countries are compared to advanced economies. In contrast, advanced economies tend to experience more significant disruptions in stock prices and capital flow volatility on average in response to rising geopolitical risks. Regional factors and underlying fundamentals may therefore play a critical role in shaping the transmission of geopolitical shocks to domestic financial markets. These findings motivate further granularity in the empirical exercise, including exploring the role of country-specific factors affecting the overall findings, as well as dynamic analysis, as described earlier.

Table 3: Geopolitical Risks, Capital Flow Volatility, and Financial Markets—Emerging Versus Advanced Economies

	Emerging Economies				Advanced Economies			
	(1) Capital Flow Volatility	(2) Stock Prices	(3) Bond Yield	(4) Exchange Rate	(5) Capital Flow Volatility	(6) Stock Prices	(7) Bond Yield	(8) Exchange Rate
GPR index	3.144*	-0.017***	0.088**	-0.010***	4.465***	-0.032***	0.037*	-0.003*
	(1.840)	(0.005)	(0.043)	(0.002)	(1.343)	(0.005)	(0.020)	(0.002)
Observations	1,569	1,970	1,758	1,970	1,610	1,998	1,998	1,998
R-squared	0.011	0.663	0.445	0.651	0.017	0.428	0.810	0.236
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Country fixed effects	YES	YES	YES	YES	YES	YES	YES	YES

GPR = geopolitical risk.

Notes: The table presents the estimated impact of geopolitical risks on capital flow volatility, stock prices, bond yields, and exchange rates, grouped by emerging and advanced economies. Control variables include domestic fundamentals, global factors, and country-specific characteristics as specified in Equation (2). Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Source: Authors' calculations.

2. Financial Development

Table 4 presents the estimated impact of geopolitical risks on capital flow volatility, stock prices, bond yields, and exchange rates based on the level of financial development. The analysis focuses on countries with high financial development (top 25th percentile) and low financial development (bottom 25th percentile). The results show that financial development slightly mitigates the effects of geopolitical risks on financial and currency markets. In countries with higher financial development, the responses of stock prices and bond yields to geopolitical risks are smaller compared to baseline estimates, while mild currency appreciation materializes potentially owing to safe-haven flows. Conversely, in countries with lower financial development, the effects on bond yields and stock prices are more pronounced. In addition, higher geopolitical risk leads to a weaker currency as foreign investors shift capital out of economies with lower levels of financial development.

These findings align with previous literature, which suggests that financial development enhances resilience to adverse financial shocks (Kalemli-Ozcan and Unsal 2023, Renzhi and Beirne 2025). However, the results also show that the impact of geopolitical risks on capital flow volatility is greater for countries with higher financial development. This may be because higher financial development often correlates with greater global financial integration, making these countries more susceptible to temporary capital flow swings during periods of heightened geopolitical risk (Magud and Pienknagura 2023).

Table 4: Geopolitical Risks, Capital Flow Volatility, and Financial Markets—Conditioning on Financial Development

	High Financial Development				Low financial development			
	(1) Capital Flow Volatility	(2) Stock Prices	(3) Bond Yield	(4) Exchange Rate	(5) Capital Flow Volatility	(6) Stock Prices	(7) Bond Yield	(8) Exchange Rate
GPR index	1.091**	-0.0242***	0.0756***	0.0131***	0.207*	-0.253***	1.259***	-0.131***
	(0.486)	(0.00514)	(0.0164)	(0.00159)	(0.105)	(0.0359)	(0.244)	(0.0142)
Observations	1,722	1,848	1,848	1,848	1,082	1,686	1,394	1,686
R-squared	0.063	0.278	0.753	0.196	0.055	0.577	0.435	0.659
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Country fixed effects	YES	YES	YES	YES	YES	YES	YES	YES

GPR = geopolitical risk.

Notes: The table presents the estimated impact of geopolitical risks on capital flow volatility, stock prices, bond yields, and exchange rates, based on the level of financial development. Control variables include domestic fundamentals, global factors, and country-specific characteristics as specified in Equation (2). Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Source: Authors' calculations.

3. Central Bank Independence

Table 5 presents the estimated impact of geopolitical risks on capital flow volatility, stock prices, bond yields, and exchange rates segmented by the level of central bank independence. The analysis distinguishes between countries with high central bank independence (top 25th percentile) and those with low central bank independence (bottom 25th percentile). The findings indicate that higher central bank independence effectively mitigates the impact of geopolitical risks on financial markets and capital flows. In countries with greater central bank independence, the adverse effects of rising geopolitical risk on capital flow volatility, stock prices, and bond yields are statistically insignificant, while mild currency depreciation impacts owing to geopolitical risks are found. In contrast, for economies with lower central bank independence, the effects of geopolitical risks are more severe. These countries experience a significant decline in stock prices and sharper currency weakening, while bond yields and capital flow volatility increase significantly in response to heightened geopolitical risks. This suggests central bank independence plays a critical role in buffering economies against the adverse effects of geopolitical uncertainty, helping stabilize financial markets and maintain investor confidence (Gadanecz, Miyajima, and Urban 2014; Caballero and Gadanecz 2023).

Table 5: Geopolitical Risks, Capital Flow Volatility, and Financial Markets—Conditioning on Central Bank Independence

	High Central Bank Independence				Low Central Bank Independence			
	(1) Capital Flow Volatility	(2) Stock Prices	(3) Bond Yield	(4) Exchange Rate	(5) Capital Flow Volatility	(6) Stock Prices	(7) Bond Yield	(8) Exchange Rate
GPR index	1.062	0.008	-0.067	-0.005**	3.604**	-0.161***	-0.142**	-0.159***
	(0.669)	(0.013)	(0.064)	(0.002)	(1.637)	(0.055)	(0.034)	(0.021)
Observations	2,116	2,632	2,632	2,632	1,407	1,311	2,116	2,632
R-squared	0.025	0.196	0.655	0.232	0.068	0.565	0.025	0.196
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Country fixed effects	YES	YES	YES	YES	YES	YES	YES	YES

GPR = geopolitical risk.

Notes: The table presents the estimated impact of geopolitical risks on capital flow volatility, stock prices, bond yields, and exchange rates, segmented by the level of central bank independence. Control variables include domestic fundamentals, global factors, and country-specific characteristics as specified in Equation (2). Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Source: Authors' calculations.

4. Public Debt

Table 6 shows the estimated impact of geopolitical risks on capital flow volatility, stock prices, bond yields, and exchange rates grouped by the public debt levels (top 25th percentile for high debt, bottom 25th percentile for low debt). The results highlight the insulating role of lower public debt levels in mitigating these risks. In economies with lower debt-to-GDP ratios, the adverse effects of heightened geopolitical risks—such as increased volatility in capital flows, stock price declines, rising bond yields, and depreciating exchange rates—are considerably weaker. On the contrary, countries with higher public debt face exacerbated impacts, experiencing more pronounced financial market disruptions, including significant declines in stock prices and currency depreciation, alongside higher bond yields and capital flow volatility. This suggests that maintaining fiscal and debt sustainability is essential for shielding economies from geopolitical and financial shocks. Our results align with broader literature, which suggests financial vulnerabilities, such as high public debt, tend to amplify the adverse effects of external shocks, while strong fiscal resilience offers some buffer against these dynamics (e.g., Beirne and Renzhi 2024).

Table 6: Geopolitical Risks, Capital Flow Volatility, and financial markets – conditioning on public debt

	High public debt				Low public debt			
	(1) Capital Flow Volatility	(2) Stock Prices	(3) Bond Yield	(4) Exchange Rate	(5) Capital Flow Volatility	(6) Stock Prices	(7) Bond Yield	(8) Exchange Rate
GPR index	10.03***	-0.0891***	0.179***	-0.012***	0.221***	-0.026***	0.090	0.003
	(1.561)	(0.0243)	(0.056)	(0.00370)	(0.0489)	(0.007)	(0.137)	(0.006)
Observations	1,343	1,427	1,427	1,427	1,343	1,495	1,378	1,495
R-squared	0.093	0.323	0.579	0.088	0.094	0.300	0.579	0.686
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Country fixed effects	YES	YES	YES	YES	YES	YES	YES	YES

GPR = geopolitical risk.

Notes: The table presents the estimated impact of geopolitical risks on capital flow volatility, stock prices, bond yields, and exchange rates, grouped by the public debt levels. Control variables include domestic fundamentals, global factors, and country-specific characteristics as specified in Equation (2). Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Source: Authors' calculations.

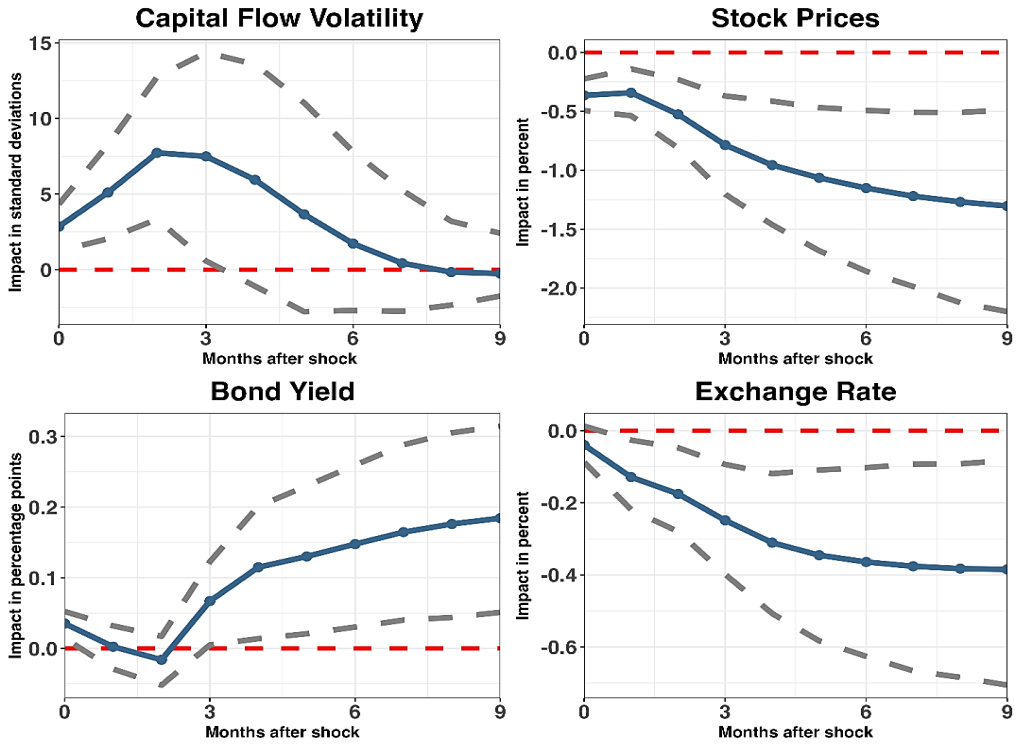
C. Dynamic Effects of Geopolitical Risks

Finally, we investigate the dynamic effects of geopolitical risks on capital flow volatility and financial market variables, using the panel SVAR model outlined in Section II.C. Figure 2 shows the estimated impulse responses based on the panel SVAR model. The solid line in each graph represents the estimated impulse responses over the following 9 months to a positive geopolitical risk shock—namely, a sudden elevation in geopolitical risk. Specifically, a geopolitical risk shock is defined as being 1 standard deviation from the GPR index. The grey dashed lines represent 95% confidence intervals generated by 500 Monte Carlo repetitions.

The results provide further evidence that heightened geopolitical risks significantly increase capital flow volatility and long-term bond yields, while stock prices fall. Currencies, meanwhile, are found to depreciate as a result of positive GPR shocks. More specifically, a 1 standard deviation rise in GPR leads to a more than proportional positive response in capital flow volatility at around 7 standard deviations after 2 months, with the response becoming insignificant after 3 months. For bond yields, these rise by around 0.1 percentage points at 3 months after the shock and rise gradually over the horizon, reaching 0.2 percentage points at 9 months after the shock. Significant negative responses are also found for stock prices, falling in the range of around 1–1.5% as a result of a 1 standard deviation positive GPR shock, with the maximum impact occurring in the range of 6 to 9 months after the shock.[6] Meanwhile, currencies depreciate by around 0.4% on average during this window.

[6] While the SVAR baseline specification and recursive ordering scheme is underpinned by economic theory, and alternative orderings are more difficult to justify based on economic theory and intuition, we consider one such scheme for robustness. Specifically, we adopt an ordering in which the GPR index follows global factors but precedes macroeconomic fundamentals, capital flow volatility, and asset market variables. The results remain qualitatively consistent with our baseline estimates (see Figure A1 in the Appendix), indicating that heightened geopolitical risks significantly increase capital flow volatility and long-term bond yields, while reducing stock prices and exchange rates.

Figure 2: Impulse Responses of Capital Flow Volatility and Financial Market Variables to a Positive Geopolitical Risk Shock



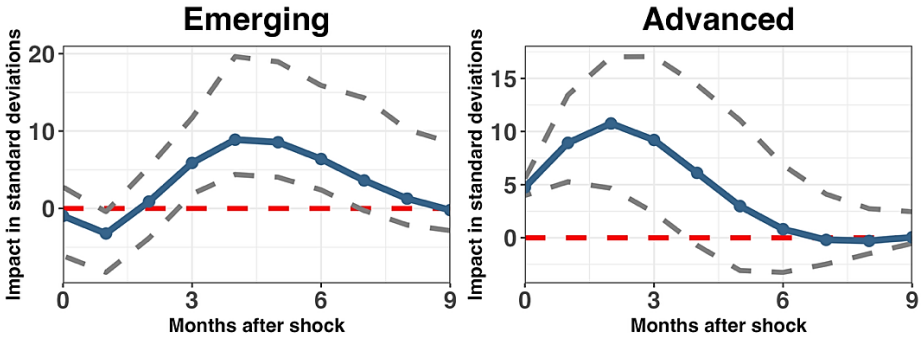
Notes: The figure plots the impulse responses of capital flow volatility and financial market variables to a positive geopolitical risk shock. The grey dashed lines represent 95% confidence intervals generated by 500 Monte Carlo repetitions.

Source: Authors' calculations.

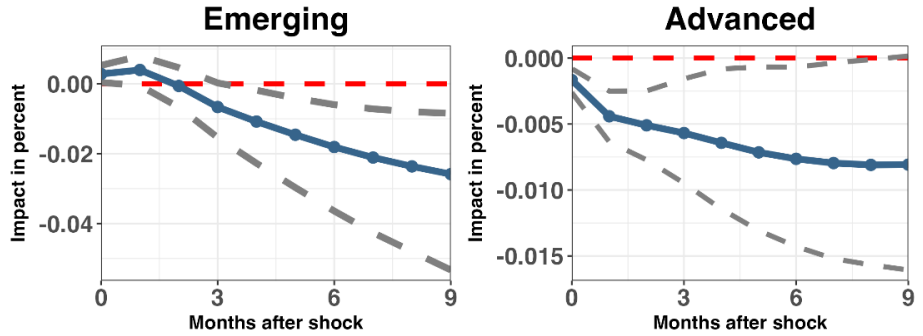
We further divide our sample into emerging and advanced economies to investigate regional heterogeneity in the dynamic effects of geopolitical risks. Figure 3 presents the estimated impulse responses, which are largely consistent with the results reported in Table 3.

Figure 3: Impulse Responses of Capital Flow Volatility and Financial Market Variables to a Positive Geopolitical Risk Shock—Emerging Versus Advanced Economies

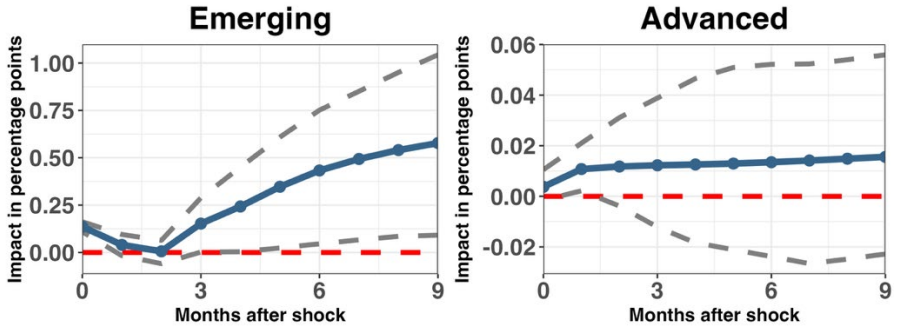
(a) Capital Flow Volatility



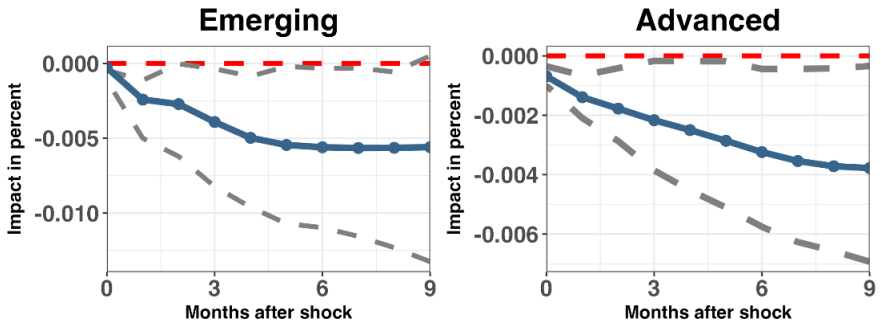
(b) Stock Prices



(c) Bond Yield



(d) Exchange Rate



Continued on the next page

Notes: The figure plots the impulse responses of capital flow volatility and financial market variables to a positive geopolitical risk shock, grouped by emerging and advanced economies. The grey dashed lines represent 95% confidence intervals generated by 500 Monte Carlo repetitions.

Source: Authors' calculations.

On capital flow volatility responses to GPR shocks, it is notable that, although the peak effect is marginally higher for advanced economies, the duration of the effect remains statistically significant for longer in the case of emerging economies (i.e., for around 6 months, which is twice as long as for advanced economies). For stock prices, while the impact of GPR on average (based on the panel OLS) had a marginally stronger effect for advanced economies, dynamic responses to GPR shocks reveal stronger negative impacts for emerging economies. Dynamic responses in bond yields are also found to be more pronounced for emerging economies. In addition, positive GPR shocks have stronger depreciating impacts on emerging market currencies compared to advanced economies, with a peak impact during the 6- to 9-month horizon after the GPR shock.

IV. CONCLUSIONS

This paper provides comprehensive insights into the significant impacts of geopolitical risks on domestic financial markets and capital flow volatility across 29 emerging and advanced economies over the period 2000 to 2023. By employing a two-stage approach that combines panel regressions and a panel SVAR model, we capture both the average and the dynamic effects of geopolitical risks on financial variables, while controlling for domestic fundamentals and global factors. For emerging economies, we find that geopolitical risks have stronger impacts on raising bond yields and greater currency depreciation effects compared to advanced economies. These findings also hold in dynamic analysis. In addition, while equities in advanced economies are slightly more negatively affected on average than emerging economies, it is also important to consider the more pronounced dynamic response in emerging equities to GPR shocks. On capital flow volatility, the average impact is slightly higher in the case of advanced economies. However, subsequent dynamic analysis reveals that the impact on capital flow volatility is much more persistent in the case of emerging economies.

Furthermore, empirical findings from the heterogeneity analysis demonstrate that country-specific characteristics, such as financial development, central bank independence, and public debt levels, play critical roles in moderating these effects. More specifically, economies with stronger financial systems and greater central bank independence are better equipped to buffer against the adverse impacts of geopolitical risks, experiencing less severe disruptions in financial markets. Conversely, countries with higher public debt or lower levels of financial development face exacerbated vulnerabilities, including sharper declines in stock prices and exchange rates, as well as increased capital flow volatility. These results underline the importance of robust macroeconomic and institutional frameworks, and financial sector structures, in mitigating the adverse effects of geopolitical risks on economic stability. This study emphasizes that, while geopolitical risks pose significant challenges to both advanced and emerging economies, ability to manage these risks is strongly influenced by the strength of a country's financial and macroeconomic fundamentals, and institutional structure. While enhancing financial development can help moderate the effects of geopolitical risks on asset markets, effective macroprudential policy and capital flow management frameworks are key aspects for consideration in helping mitigate capital flow volatility.

Geopolitical risk spillovers to domestic financial markets carry significant implications for monetary policy and central banks, especially as they heighten economic uncertainty and erode investor and consumer confidence. Such dynamics can influence inflation trajectories over the forecast horizon, complicating monetary policy decisions and potentially weakening the effectiveness of policy transmission mechanisms. As well as strengthening central bank independence, geopolitical risks also warrant consideration by central banks and financial supervisors from a financial stability perspective. Notably, our findings underscore the importance for central banks of closely monitoring how geopolitical risks spill over into asset markets. These findings are especially relevant for policymakers looking to strengthen the resilience of domestic financial markets and capital flows to external shocks driven by rising geopolitical tensions and associated economic uncertainties. Our results offer valuable insights into how monetary and macroprudential policies could be leveraged to mitigate these risks. In addition, policymakers should strengthen the coordination between monetary and macroprudential policies, which can bolster the response of monetary and financial authorities with regard to stabilizing markets during episodes of elevated geopolitical risk.

APPENDIX

Table A1: Panel Unit Root Test

Variables	Test Statistics	p-value
GPR index	-14.82	0.000
Capital flow volatility	-15.872	0.000
Real stock prices	-3.93	0.000
Long-term bond yield	-4.03	0.000
Real effective exchange rate	-2.88	0.002
Policy rate	-8.47	0.000
Real industrial production	-5.89	0.000
Inflation	-16.53	0.000
Oil price	-9.65	0.000
VIX	-13.31	0.000
EPU index	-7.77	0.000
Financial development index	-9.89	0.000
Central bank independence index	-10.11	0.000
Public debt to GDP	-4.34	0.000

EPU = Economic Policy Uncertainty, GDP = gross domestic product, GPR = geopolitical risk, VIX = Chicago Board Options Exchange Volatility Index.

Notes: The table presents the result of the panel unit root test, conducted following Choi (2001), incorporating corrections for cross-sectional dependence. The lag length for the augmented Dickey–Fuller regressions is set to 10.

Source: Authors' calculations.

Table A2: Geopolitical Risks and Capital Flow Volatility—Alternative Capital Flow Volatility Measures

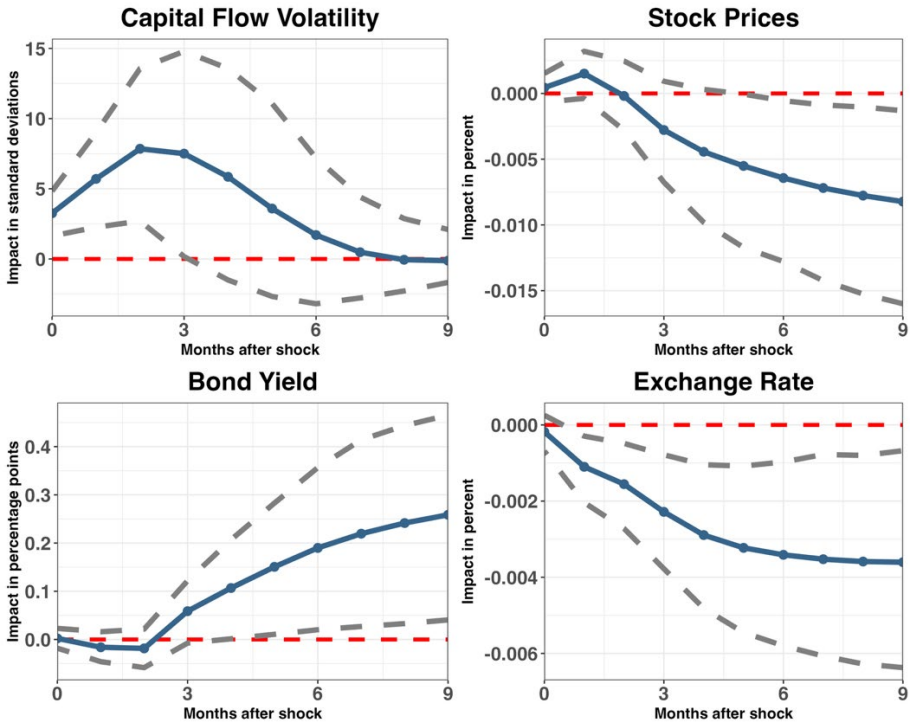
	(1) GARCH	(2) ARIMA	(3) RW: 6	(4) RW: 12
GPR index	8.607***	4.126**	7.527***	5.263***
	(0.805)	(1.754)	(0.824)	(0.875)
Observations	3,968	3,968	3,179	3,179
R-squared	0.880	0.027	0.059	0.089
Controls	YES	YES	YES	YES
Country fixed effects	YES	YES	YES	YES

GPR = geopolitical risk.

Notes: The table presents the estimated impact of geopolitical risks on capital flow volatility, using alternative capital flow volatility measures. Control variables include domestic fundamentals, global factors, and country-specific characteristics as specified in Equation (2). Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Source: Authors' calculations.

Figure A1: Impulse Responses of Capital Flow Volatility and Financial Market Variables to a Positive Geopolitical Risk Shock—Alternative Ordering



Notes: The figure plots the impulse responses of capital flow volatility and financial market variables to a positive geopolitical risk shock, based on alternative ordering in which the geopolitical risk index follows global factors but precedes macroeconomic fundamentals, capital flow volatility, and asset market variables. The grey dashed lines represent 95% confidence intervals generated by 500 Monte Carlo repetitions.

Source: Authors' calculations.

REFERENCES

Abdel-Latif, Hany, and Mahmoud El-Gamal. 2020. "Financial Liquidity, Geopolitics, and Oil Prices." *Energy Economics* 87: 104482. https://doi.org/10.1016/j.eneco.2019.104482

Apergis, Nicholas, Matteo Bonato, Rangan Gupta, and Clement Kyei. 2018. "Does Geopolitical Risks Predict Stock Returns and Volatility of Leading Defense Companies? Evidence from a Nonparametric Approach." *Defense Peace Economics* 29(6): 684–696.

Baker, Scott, Nicholas Bloom, and Stephen Davis. 2016. "Measuring Economic Policy Uncertainty." *The Quarterly Journal of Economics* 131(4): 1593–1636.

Balcilar, Mehmet, Matteo Bonato, Riza Demirer, and Rangan Gupta. 2018. "Geopolitical Risks and Stock Market Dynamics of the BRICs." *Economic Systems* 42(2): 295–306.

Banerjee, Ameet, Ahmet Kumar, and John Goodell 2024. "Volatility Connectedness Between Geopolitical Risk and Financial Markets: Insights from Pandemic and Military Crisis Periods." *International Review of Economics and Finance* 96: 103740. https://doi.org/10.1016/j.iref.2024.103740

Banerjee, Debanjan, Arjit Ghosal, and Imon Mukherjee. 2019. "Prediction of Gold Price Movement Using Geopolitical Risk as a Factor." In Ajith Abraham, Paramartha Dutta, Jyotsna Mandal, Abhishek Bhattacharya, and Soumi Dutta. eds. *Emerging Technologies in Data Mining and Information Security. Advances in Intelligent Systems and Computing* (pp. 879–886). Singapore: Springer.

Baur, Dirk, and Lee Smales. 2020. "Hedging Geopolitical Risk with Precious Metals." *Journal of Banking & Finance* 117: 105823. https://doi.org/10.1016/j.jbankfin.2020.105823

Behera, Chinmaya, Biswashree Priyadarsini, and Debasis Patnaik. 2024. "Impact of Geopolitical Risk and Crude Oil Prices on Stock Return." *Bulletin of Monetary Economics and Banking* 27(1): 45–58.

Beirne, John, and Nuobu Renzhi. 2024. "Debt Shocks and the Dynamics of Output and Inflation in Emerging Economies." *Journal of International Money and Finance* 148: 103167. https://doi.org/10.1016/j.jimonfin.2024.103167

Broto, Carmen, Javier Díaz-Cassou, and Aytor Erce. 2011. "Measuring and Explaining the Volatility of Capital Flows to Emerging Countries." *Journal of Banking & Finance* 35(8): 1941–1953.

Caballero, Julián, and Blaise Gadanecz. 2023. "Did Interest Rate Guidance in Emerging Markets Work?" BIS Working Paper 1080.

Caldara, Dario, and Matteo Iacoviello. 2022. "Measuring Geopolitical Risk." *American Economic Review* 112(4): 1194–1225.

Choi, In. 2001. "Unit Root Tests for Panel Data." *Journal of International Money and Finance* 20(2): 249–272.

Christiano, Lawrence, Martin Eichenbaum, and Charles Evans. 1999. "Monetary Policy Shocks: What Have We Learned and to What End?" *Handbook of Macroeconomics* 1: 65–148.

Das, Debojyoti, Manoharan Kannadhasan, and Malay Bhattacharyya. 2019. "Do the Emerging Stock Markets React to International Economic Policy Uncertainty, Geopolitical Risk and Financial Stress Alike?" *North American Journal of Economics and Finance* 48: 1–19.

De Crescenzio, Annamaria, and Etinne Lepers. 2024. “Extreme Capital Flow Episodes from the Global Financial Crisis to COVID-19: An Exploration with Monthly Data.” *Open Economies Review* 1–32.

Demiralay, Sercan, and Erhan Kilincarslan. 2019. “The Impact of Geopolitical Risks on Travel and Leisure Stocks.” *Tourism Management* 75: 460–476.

Gadanecz, Blaise, Ken Miyajima, and Jörg Urban. 2014. “How Might EME Central Banks Respond to the Influence of Global Monetary Factors?” BIS Paper 78c.

Gkillas, Konstantinos, Rangan Gupta, and Christian Pierdzioch. 2020. “Forecasting Realized Gold Volatility: Is There a Role of Geopolitical Risks?” *Finance Research Letters* 35: 101280. https://doi.org/10.1016/j.frl.2019.08.028

Hachicha, Fatma. 2023. “Sentiment Investor, Exchange Rates, Geopolitical Risk and Developing Stock Market: Evidence of Co-Movements in the Time-Frequency Domain During Russia Ukraine War.” *Review of Behavioral Finance* 16(3): 486–509.

Hoque, Mohammad, and Mohd Zaidi. 2020. “Global and Country-Specific Geopolitical Risk Uncertainty and Stock Return of Fragile Emerging Economies.” *Borsa Istanbul Review* 20(3): 197–213.

Izzeldin, Marwan, Yaz Muradoğlu, Vasileios Pappas, Athina Petropoulou, and Sheeja Sivaprasad. 2023. “The Impact of the Russian-Ukrainian War on Global Financial Markets.” *International Review of Financial Analysis* 87: 102598. https://doi.org/10.1016/j.irfa.2023.102598

Jiang, Yonghon, Gengyu Tian, Yiqi Wu, and Ben Mo. 2020. “Impacts of Geopolitical Risks and Economic Policy Uncertainty on Chinese Tourism-Listed Company Stock.” *International Journal of Finance and Economics* 27(1): 320–333.

Kalemli-Ozcan, Şebnem, and Filiz Unsal. 2023. “Global Transmission of Fed Hikes: The Role of Policy Credibility and Balance Sheets.” Brookings Papers on Economic Activity.

Liu, Jing, Feng Ma, Yingkai Tang, and Yaojie Zhang. 2019. “Geopolitical Risk and Oil Volatility: A New Insight.” *Energy Economics* 84:104548. https://doi.org/10.1016/j.eneco.2019.104548

Ma, Shengnang. 2025. “Geopolitical Risk and Exchange Rate Returns in Asia.” *Emerging Markets Finance and Trade*, forthcoming

Magud, Nicolas, Samuel Pienknagura. 2023. “External Shocks, Policies, and Tail-Shifts in Real Exchange Rates.” IMF Working Paper 2023(129).

Mansour-Ichrakieh, Layal, and Hussein Zeaiter. 2019. “The Role of Geopolitical Risks on the Turkish Economy Opportunity or Threat.” *North American Journal of Economics and Finance* 50(9): 101000. https://doi.org/10.1016/j.najef.2019.101000

Mei, Dexiang, Feng Ma, Yin Liao, and Lu Wang. 2020. “Geopolitical Risk Uncertainty and Oil Future Volatility: Evidence from MIDAS Models.” *Energy Economics* 86: 104624. https://doi.org/10.1016/j.eneco.2019.104624

Qin, Yun, Kairong Hong, Jinyu Chen, and Zitao Zhang. 2020. “Asymmetric Effects of Geopolitical Risks on Energy Returns and Volatility Under Different Market Conditions.” *Energy Economics* 90: 104851. https://doi.org/10.1016/j.eneco.2020.104851

Renzhi, Nuobu, and John Beirne. 2025. “Global Shocks and Monetary Policy Transmission in Emerging Markets.” *Emerging Markets Finance and Trade* 61: 786–803.

Romelli, Davide. 2022. “The Political Economy of Reforms in Central Bank Design: Evidence from a New Dataset.” *Economic Policy* 37(112): 641–688.

Saiz, Albert, and Uri Simonsohn. 2013. “Proxying for Unobservable Variables with Internet Document-Frequency.” *Journal of the European Economic Association* 11(1): 137–165.

Salisu, Afees, Ahemuefula Ogbonna, Lukman Lasisi, and Abeeb Olaniran. 2022. “Geopolitical Risk and Stock Market Volatility in Emerging Markets: A GARCH–MIDAS Approach.” *The North American Journal of Economics and Finance* 62: 101755.

Sharpe, William. 1964. “Capital Asset Prices: A Theory of Market Equilibrium Under Conditions of Risk.” *Journal of Finance* 19(3): 425–442.

Smales, Lee. 2021. “Geopolitical Risk and Volatility Spillovers in Oil and Stock Markets.” *The Quarterly Review of Economics and Finance* 80: 358–366.

Wang, Kai-Hua, Chi-Wei Su, and Muhammad Umar. 2021. “Geopolitical Risk and Crude Oil Security: A Chinese Perspective.” *Energy* 219: 119555. https://doi.org/10.1016/j.energy.2020.119555

Yilmazkuday, Hakan. 2024. “Geopolitical Risk and Stock Prices.” *European Journal of Political Economy* 83: 102553.

Zhang, Du, and Rundong Chen. 2025. “The Impact of Economic Uncertainty on Cross-Border Capital Flows.” *Emerging Markets Finance and Trade* 61(11): 3423–3445.

Zhang, Yaojie, Jiaxin He, Mengxi He, and Shaofang Li. 2023. “Geopolitical Risk and Stock Market Volatility: A Global Perspective.” *Finance Research Letters* 53: 103620. https://doi.org/10.1016/j.frl.2022.103620

Geopolitical Risk, Capital Flow Volatility, and Asset Market Spillovers

This paper explores how country-specific geopolitical risks affect capital flow volatility and asset markets across 29 advanced and emerging economies from 2000 to 2023. While these risks have negative spillovers across all economies, emerging economies are especially impacted through bond yields and exchange rates. The paper also shows that deeper financial development, higher central bank independence, and lower public debt can help to mitigate the effects of geopolitical risks and strengthen resilience.

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